



PHYSICS

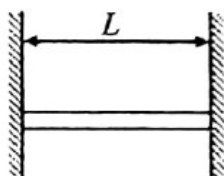
Max Marks : 60

Section-1

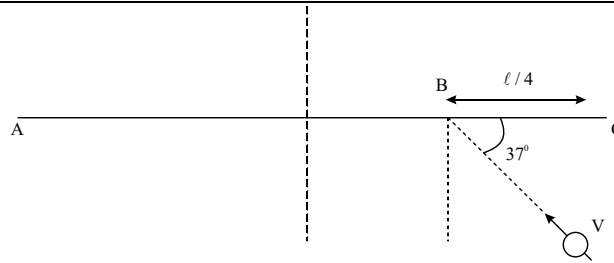
(One or more options correct type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D).
Out of Which Only One is correct

1. A horizontal rod (length L , mass m , cross section A , coefficient of linear expansion α) fits exactly (with out tension) between two rigid vertical supports choose the correct option

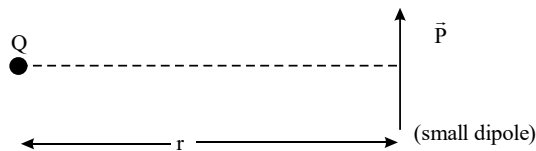


- A) Since the length of rod cannot increase beyond L , on increasing temperature, the effect of thermal expansion is neutralized by compressive forces developed by the wall
- B) If the walls and rod are rough with mutual friction coefficient μ , then increment in temperature θ of the rod, required to prevent the rod from falling is
- $$\Delta\theta = \frac{mg}{2\mu YA\alpha}$$
- C) If the walls and rod are rough with mutual friction coefficient μ , then increment in temperature θ of the rod, required to prevent the rod from falling is $\Delta\theta = \frac{mg}{4\mu YA\alpha}$
- D) The stress developed in wire is $\alpha y \Delta\theta$
2. A uniform rod AC of length l and mass m is kept on a horizontal smooth plane. It is free to rotate and move. A particle of same mass m moving on the plane with velocity v strike the rod at point B making an angle 37° with the rod. The collision is elastic. After collision.



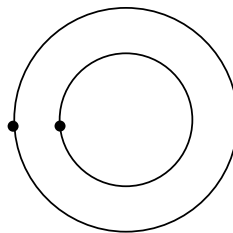
- A) The angular velocity of the rod will be $\frac{72v}{55l}$
- B) The centre of the rod will travel a distance $\frac{\pi l}{3}$ in the time in which it makes half rotation
- C) Impulse of the impact force is $\frac{24mv}{55}$
- D) All the above are correct

3. For the situation shown in the figure (Assume $r \gg$ length of dipole) mark out the correct statements.



- A) Force acting on the dipole is zero
- B) Force acting on the dipole is $\frac{PQ}{4\pi\epsilon_0 r^3}$
- C) Torque acting on the dipole is $\frac{2PQ}{4\pi\epsilon_0 r^2}$
- D) Torque acting on the dipole is $\frac{PQ}{2\pi\epsilon_0 r^2}$

- 4) Two concentric, coplanar, circular loop of wire, with different diameter carry current in the same sense as shown in the figure. Which of the following statement is correct?



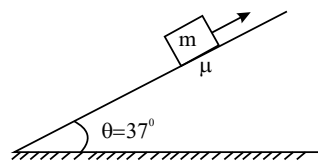
- A) The magnetic force exerted by the outer loop on a short portion of the inner loop is radially outward
- B) The magnetic force exerted by the outer loop on a short portion of the inner is radially outward

C) The net magnetic force exerted by the outer loop on a whole inner is non zero and is radially outward

D) The net magnetic force exerted by the outer loop on a whole inner loop is zero

- 5). A block of mass m is being pulled up the rough incline, inclined at an angle 37° with horizontal by an agent delivering constant power P . The coefficient of friction between the block and the incline is μ . Find the maximum speed (in m/s) of the block during the course of ascent.

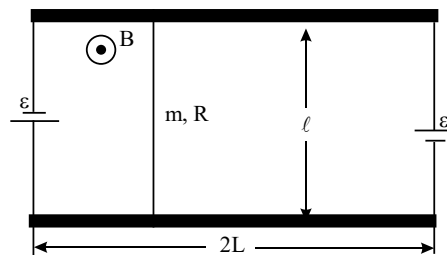
[Take : $P = 60\text{W}$, $m = 1\text{kg}$, $\mu = 0.5$]



- A) 1 B) 3 C) 5 D) 6

6. Side rail of length $2L$ are fixed on a horizontal plane at a distance l from each other. These ends are connected by two identical ideal batteries with emf E by resistanceless wires (see figure). On the rails is a rod of mass m , which may slide along them. The entire system is placed in a uniform vertical magnetic field B .

Assuming that the resistance of the rod is R and the resistance per unit length of each of the rails equal to ρ , find the period of small oscillations (in sec.) arising from shifting the rod from the equilibrium along the rails. Neglect friction, internal resistance of batteries and induced emf in the rod. [Take : $B = \pi T$, $\varepsilon = \pi$ volt, $l = 0.5m$, $L = 1m$, $\rho = 1\Omega/m$, $R = 0.25\Omega$, $m = 100\text{gm}$]

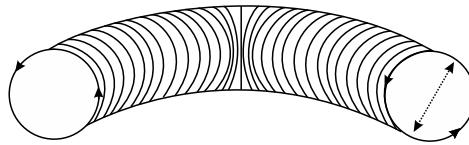


- A) 1 B) 3 C) 5 D) 6

7. A wooden plank of length 1m and uniform cross section is hinged at one end to the bottom of a tank as shown in the figure. The tank is filled with water up to a height of 0.5m. The specific gravity of the plank is 0.5. θ is angle which plank makes with the vertical in the equilibrium position (exclude the case $\theta = 0$). Find the value of $\frac{1}{\cos^2 \theta}$.

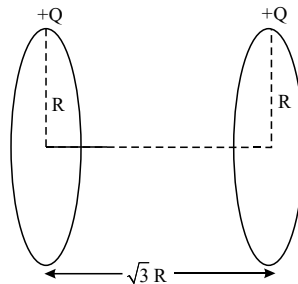
A) 1 B) 2 C) 3 D) 4

8. Calculate the magnetic moment (in Am^2) of a thin wire with a current $I = 8.0\text{A}$, wound tightly on a half a toroid (see figure). The diameter of the cross section of the toroid is equal to $d = 5.0\text{cm}$, and the number of turns is $N = 500$.



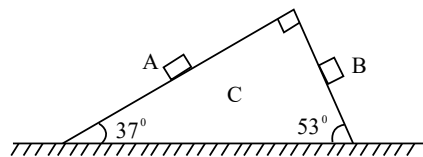
A) 5 B) 6 C) 7 D) 8

9. Two identical charged rings of charge $Q=1\text{C}$ and radius $R = 0.1\text{m}$ are separated by a distance $\sqrt{3}R$ meter, and are fixed. A point charge of charge $q = 1.0 \times 10^{-9}\text{C}$ and mass $m = 2\text{kg}$ is moved by external agent from the centre of left ring to the centre of right ring. Find the work done by an external agent in joule.



A) 0 B) 1 C) 2 D) 3

- 10). In the figure shown blocks 'A' and 'B' are kept on a wedge 'C'. A, B and C have mass m . All surfaces are smooth. Find the acceleration of C.



A) 0 B) 1 C) 2 D) 3

SECTION - II

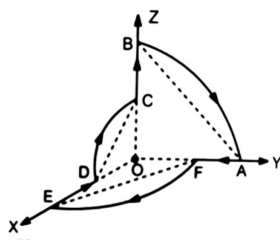
(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS** correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Paragraph for Question Nos. 11 & 12

In a rigid wire frame ABCDEFA, as shown in the figure, a current I_0 flows through the wire from A to F as shown in the figure. Radii of the three quarters AB, CD and EF are $r_1 = 3^{\frac{1}{4}}m$, $r_2 = 3^{\frac{-1}{4}}m$ and $r_3 = 1m$ respectively with common centre at O. All the wires are light except for the portion ED which have a mass of 'm' kg.



11. The magnetic moment of the loop is

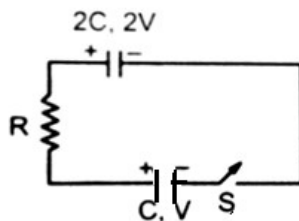
- A) $\frac{\pi I_0}{4} \left[\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$ B) $-\frac{\pi I_0}{4} \left[\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$
 C) $\frac{\pi I_0}{4} \left[-\sqrt{3}\hat{i} - \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$ D) $\frac{\pi I_0}{4} \left[-\sqrt{3}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \hat{k} \right]$

12. If this loop is kept in a magnetic field of magnetic 2T which is directed along positive y-axis, angular acceleration of the loop is: (where l =length of portion ED)

- A) $\frac{8\pi I_0}{ml^2} [\hat{i} + \sqrt{3}\hat{k}]$ B) $\frac{6\pi I_0}{ml^2} [\hat{i} + \sqrt{3}\hat{k}]$ C) $\frac{8\pi I_0}{ml^2} [\hat{i} - \sqrt{3}\hat{k}]$ D) $\frac{6\pi I_0}{ml^2} [\hat{i} - \sqrt{3}\hat{k}]$

Paragraph for Question Nos. 13 & 14

Two capacitors of capacitance $2C$ and C are charged to potential difference $2V$ and V respectively. Now the charged capacitors are connected with a resistor R as shown in figure. Switch 'S' is closed at $t=0$.



13. What is the total heat produced in circuit till steady state is obtained?

- A) $\frac{1}{3}CV^2$ B) $\frac{CV^2}{2}$ C) $\frac{2}{3}CV^2$ D) $\frac{3}{5}CV^2$

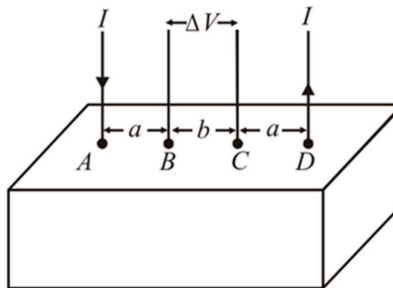
14. What is the current in circuit as function of time?

- A) $\frac{V}{R}e^{-3t/RC}$ B) $\frac{2V}{R}e^{-2t/3RC}$ C) $\frac{V}{3R}e^{-t/2RC}$ D) $\frac{V}{R}e^{-3t/2RC}$

Paragraph for Question Nos. 15 & 16

Consider a block of conducting material of resistivity ρ shown in the figure. Current I enters at A and leaves from D. We apply superposition principle to find voltage ΔV developed between B and C. The calculation is done in the following steps:

- Take current I entering from A and assume it to spread over a hemispherical surface in the block.
- Calculate field $E(r)$ at distance r from A by using Ohm's law $E = \rho j$, where j is the current per unit area at r .
- From the r dependence of $E(r)$, obtain the potential $V(r)$ at r .
- Repeat (i), (ii) and (iii) for current I leaving D and superpose results for A and D.



15. ΔV measured between B and C is

- A) $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$ B) $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$ C) $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$ D) $\frac{\rho I}{2\pi(a-b)}$

16. For current entering at A, the electric field at a distance ' r ' from A is

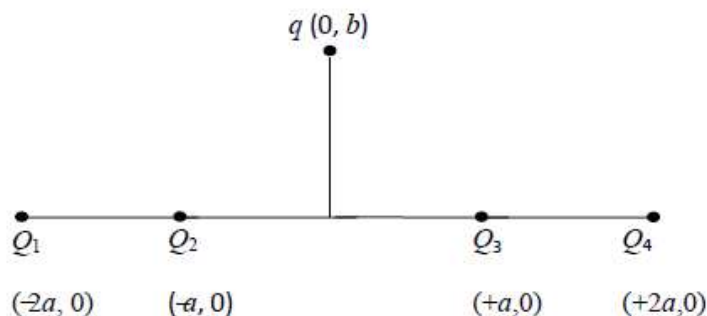
- A) $\frac{\rho I}{8\pi r^2}$ B) $\frac{\rho I}{\pi r^2}$ C) $\frac{\rho I}{2\pi r^2}$ D) $\frac{\rho I}{4\pi r^2}$

Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (**List-1** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.



LIST-I		LIST-II	
P	Q_1, Q_2, Q_3, Q_4 all positive	1	+x
Q	Q_1, Q_2 positive: Q_3, Q_4 negative	2	-x
R	Q_1, Q_4 positive: Q_2, Q_3 negative	3	+y
S	Q_1, Q_3 positive: Q_2, Q_4 negative	4	-y

A) P-3, Q-1, R-4, S-2

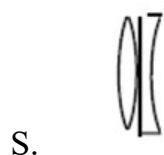
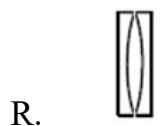
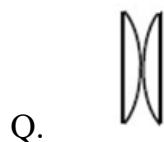
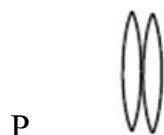
B) P-4, Q-2, R-3, S-1

C) P-3, Q-1, R-2, S-4

D) P-4, Q-2, R-1, S-3

18. Four combinations of two thin lenses are given in List I. The radius of curvature of all curved surfaces is r and the refractive index on all the lenses is 1.5. Match lens combinations in List I with their focal length in List II and select the correct answer using the code given below the lists.

List I



List II

1. $2r$

2. $r/2$

3. $-r$

4. r

Choices :

A) P-1, Q-2, R-3, S-4

B) P-2, Q-4, R-3, S-1

C) P-4, Q-1, R-2, S-3

D) P-2, Q-1, R-3, S-4

19. A block of mass $m_1 = 1\text{ kg}$ another mass $m_2 = 2\text{ kg}$, are placed together (see figure) on an inclined plane with angle of inclination θ . Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g .

[Useful information : $\tan(55^\circ) \approx 0.1$; $\tan(11.5^\circ) \approx 0.2$; $\tan(16.5^\circ) \approx 0.3$]

List I

P. $\theta = 5^\circ$

Q. $\theta = 10^\circ$

R. $\theta = 15^\circ$

S. $\theta = 20^\circ$

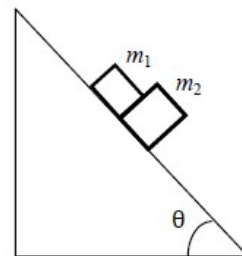
List II

1. $m_2 g \sin \theta$

2. $(m_1 + m_2) g \sin \theta$

3. $\mu m_2 g \cos \theta$

4. $\mu(m_1 + m_2) g \cos \theta$



Choices :

- A) P-1, Q-1, R-1, S-3 B) P-2, Q-2, R-2, S-3
C) P-2, Q-2, R-2, S-4 D) P-2, Q-2, R-3, S-3

20. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance of 1.2 m from the person. In the following state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given List II. Match the statements from List I with those in List II and select the correct answer using the code given below the list

List I

- P. Lift is accelerating vertically up
Q. Lift is accelerating vertically down with an acceleration less than the gravitational acceleration
R. Lift moving vertically up with constant speed
S. Lift is falling freely

List II

1. $d = 1.2m$
2. $d > 1.2m$
3. $d < 1.2m$
4. No water leaks out of the jar

- A) P-2, Q-3, R-2, S-4 B) P-2, Q-3, R-1, S-4
C) P-1, Q-1, R-1, S-4 D) P-2, Q-3, R-1, S-1

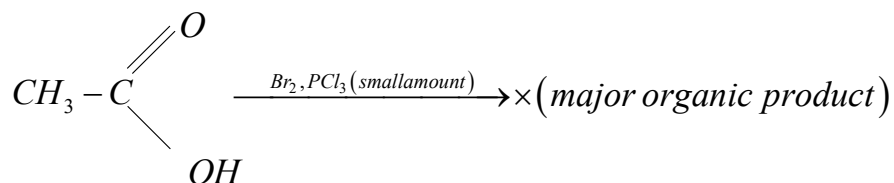
Section-1

(One or more options correct type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D).

Out of Which Only One is correct

21.



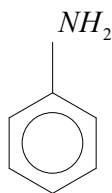
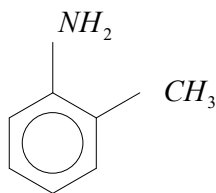
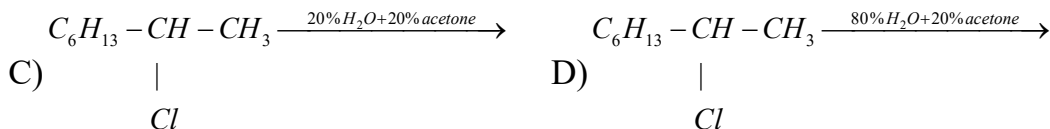
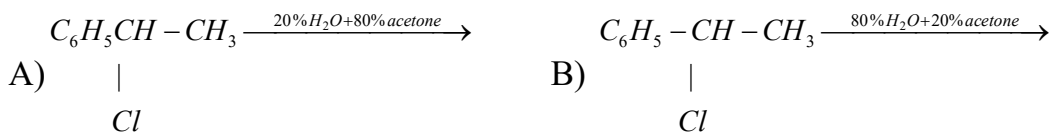
Incorrect statement among the following is

'X' is $\text{CH}_2 - \text{COOH}$ (B) In the formation of 'X' CH_3COCl intermediate involved

(C) In the formation of 'X' enolate intermediate involved

(D) $\text{Br}_2 + \text{Red P}_4$ can also be used to get 'X'

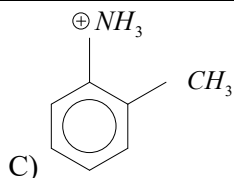
22. Among the following reaction in which one percentage of racemization is highest



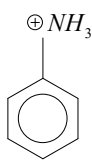
23. is less basic than in aqueous solution the main reason for it is

A) Here $-\text{CH}_3$ acts as electron withdrawing group

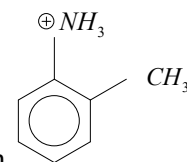
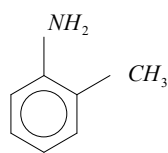
B) Steric inhibition of resonance in



is less stable than



, due to steric inhibition of hydration in

D) Due to crowding from $-CH_3$,

can not accept proton easily when compared with aniline

24. Heat of combustion of n-Hexane, n-heptane and n-octane are -1002.4 k cal/mol, -1159.9 k cal /mol and -1317.2 k cal/mol respectively. If heat of combustion of 2- methylpentane is -990 k cal/mole, heat of combustion of 2- methylnonane is nearly
- A) -1618 k.cal/mol B) -1650 k cal/mol
C) -1550 k cal/mole D) -1700 k cal/mole
25. Absolute value of which of the following thermodynamic property of a substance can be calculated?
- A) Internal energy B) Enthalpy
C) Gibbs energy D) Entropy
26. Liquid A and liquid B are immiscible liquids. At $25^{\circ}C$ vapour pressures of pure liquid A and pure liquid B are 100 mm of Hg and 200 mm of Hg respectively. Total vapour pressure of mixture of 2 moles liquid A and 3 moles of liquid B at $25^{\circ}C$ is
- A) 160 mm of Hg B) 200 mm of Hg C) 250 mm of Hg D) 300 mm of Hg
27. At 298 k the standard free energy of formation of $H_2O_{(l)}$ is -237.23 kJ/mole, standard free energy change for ionization $H_2O_{(l)}$ to $H^+_{(aq)}$ and $OH^-_{(aq)}$ is +79.71 kJ/mole. The reversible EMF at 298 k of the cell is $Pt; H_2(1bar) | H^+(1M) || OH^-(1M) | O_2(1bar); Pt$
- A) +0.403V B) +0.806 V C) +0.2 V D) +1.0V
28. EAN of iron in $Fe_2(CO)_9$ is
- A) 34 B) 35 C) 36 D) 37

29. When molten ICl subjected to electrolysis with inert electrodes, anode product is/are
 A) I_2 only B) Cl_2 only C) Both I_2 and Cl_2 D) ICl_3
30. Among the following halides for which one ΔH_f° more negative
 A) LiF B) NaF C) RbF D) CsF

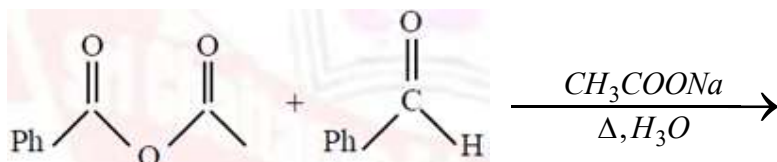
Section-2

(Paragraph Type)

This section contains 3 paragraphs each describing theory, experiment, data etc. Six questions relate to three paragraphs with two questions on each paragraph. Each question pertaining to a particular paragraph should have **only one correct answer** among the four choices A, B, C and D.

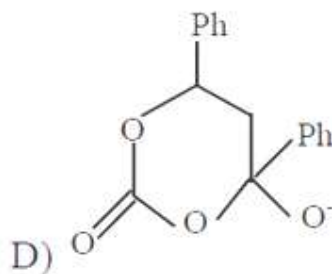
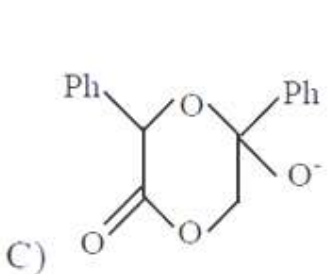
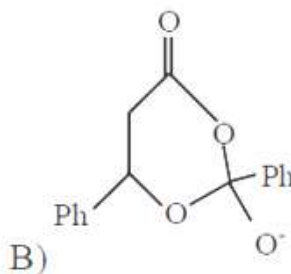
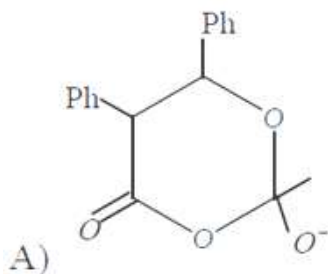
Paragraph for Questions 31 & 32

Condensation reaction between aromatic aldehyde and aliphatic acid anhydride in presence of sodium or potassium salt of the acid corresponding to the anhydride yields α, β -unsaturated aromatic acid

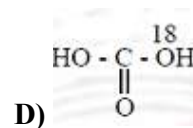
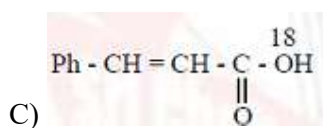
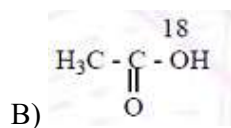
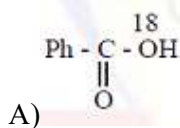


(Observe the above reaction)

31. The intermediate product in the above reaction is



32. If $\text{Ph}-\text{CH}=\text{O}^{18}$ is used instead of $\text{PhCH}=\text{O}$ in original reaction the " O^{18} " presence shall be observed in which product formed (assuming released water molecule in course of reaction involves in hydrolysis in priority at final step)



Paragraph for Questions 33 & 34

Imagine an atom made up of a proton and a hypothetical particle of half the mass of the electron but having the same charge as the electron. Bohr's model is applicable and all transition of this hypothetical particle from higher level to ground state is possible.

33. The largest wavelength of photon that will be emitted when hypothetical particle jumps to first excited level is
- (A) $\frac{18}{5R}$ (B) $\frac{72}{5R}$ (C) $\frac{36}{5R}$ (D) $\frac{8}{R}$
34. The velocity of this particle in second Bohr orbit is
- (A) $2.188 \times 10^6 \text{ m/sec}$ (B) $1.094 \times 10^6 \text{ m/sec}$ (C) $0.547 \times 10^6 \text{ m/sec}$ (D) $4.376 \times 10^6 \text{ m/sec}$

Paragraph for Questions 35 & 36

A chemist opened a cupboard to find four bottles containing water solutions, each of which has lost its label. Bottle 1, 2 and 3 contained colourless solutions, while bottle 4 contained a blue solution. The labels from the bottles were lying scattered on the floor of the cupboard. They were:

Copper (II) sulphate, Hydrochloric acid, Lead nitrate Sodium carbonate

By mixing samples of the contents of the bottles, in pairs, the chemist made the following observation:

Bottle 1 + Bottle 2	White precipitate	Bottle 1 + Bottle 3	White precipitate
Bottle 1 + Bottle 4	White precipitate	Bottle 2 + Bottle 3	Colourless gas evolved
Bottle 2 + Bottle 4	No visible reaction	Bottle 3 + Bottle 4	Blue precipitate

35. Bottle 3 contains
- A) copper (II) sulphate B) hydrochloric acid
C) lead nitrate D) sodium carbonate
36. When bottle 1 is mixed with bottle 4, white precipitate is observed, which is
- A) PbSO_4 B) CuCO_3 C) PbCl_2 D) CuCl_2

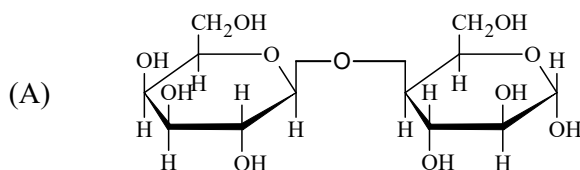
Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (**List-1** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

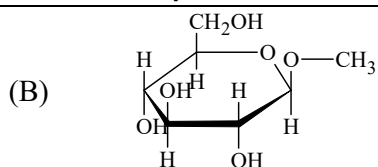
37. Match the Column:

Column – I



Column – II

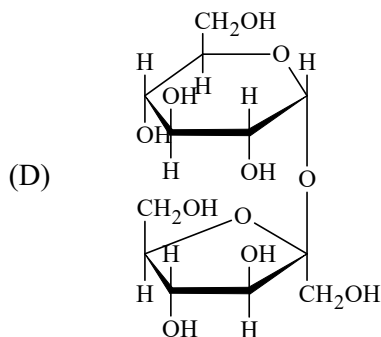
(p) glycosidic linkage



(q) reducing sugar

(C) Starch

(r) oligosaccharide



(s) non-sugar

The correct match is

(A) A-PQR,B-PQR,C-PS,D-P

(B) A-PQ,B-PQR,C-PS,D-PR

(C) A-PQR,B-PQ,C-P,D-PR

(D) A-PQR,B-P,C-PS,D-PR

38. **Column – I****Column – II****(Arrangement in unit cell, radius ratio in higher limit)****(Coordination number of cation : Anion)**

A) Cations in CCP and anions in alternate tetrahedral voids

P) ratio of number of cation to anion in one unit cell is 1 : 1

B) Cations in simple cubic and anions in the body centre.

Q) ratio of coordination number of cation to anion is 1 : 1

C) Anions in CCP and cations in all tetrahedral voids.

R) number of next neighbours of ion is greater than 10.

D) Cations in CCP and anions in all octahedral voids. S) Effective number of formula units = 4

The correct match is

A) A – P, Q,S,; B – P,Q,R,S; C – P,S,; D – P,Q,S,

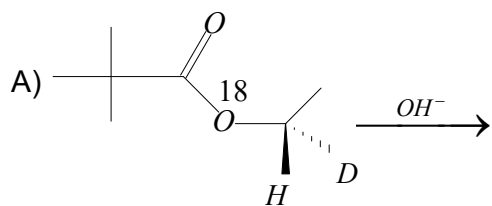
B) A – P,Q; B – P,Q,R,S; C – S,; D – P,Q,S,

C) A – P,Q,R,S,; B – P,Q,; C – R,S; D – P,Q,R,S

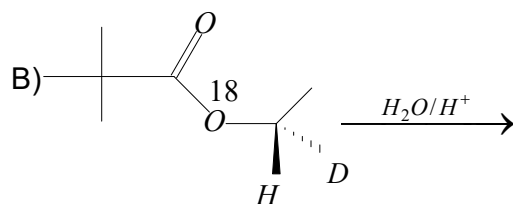
D) A – P,Q,S,; B – P,Q,R; C – P,Q,S,; D – S

39. Match the column:

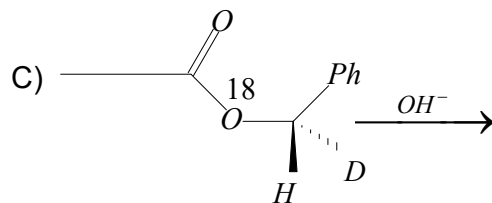
Column – I



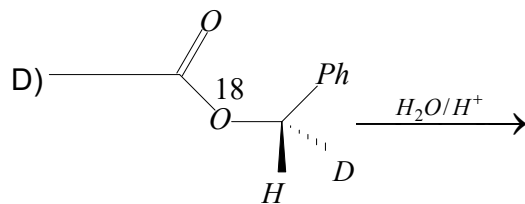
P) Bimolecular path



Q) Racemic mixture of alcohol



R) Acyl oxygen cleavage

S) Alcohol with O^{18}

T) Inversion (only) of configuration in alcohol part

The correct match is

A) A-P,R,S; B-P,R,S; C-Q,T; D-QT

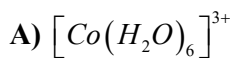
B) A-R,S ; B-P,T; C-Q ; D-P,R

C) A-P,T; B- R,S; C-P,R,S; D-Q

D) A-P,S; B-R,S; C-Q; D-P,R

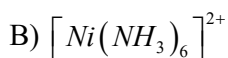
40. Match the column:

Column – I

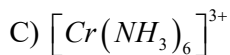


Column – II

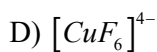
P) Low spin complex



Q) Paramagnetic complex



R) Inner orbital complex



S) Two bonds (metal-ligand) are longer than the remaining four bonds

The correct match is

A) A-P,R; B-P,Q,R; C-P,Q,R; D- QS

B) A-P,R; B-Q; C-Q,R; D-Q,S

C) A-Q,S; B-Q; C-Q,R; D-Q,S

D) A-Q,S; B-P,R,S; C-Q,R; D-Q

MATHEMATICS

Max Marks : 60

Section-1

(Only one Option correct Type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct

41. If 6, 8 and 12 are $l^{\text{th}}, m^{\text{th}}$ and n^{th} terms of an A.P. and $f(x) = nx^2 + 2lx - 2m$, then the equation

$f(x) = 0$ has

A) a root between 0 and 1

B) both roots imaginary

C) both roots negative

D) both roots greater

42. If $f(x) = \frac{1}{2} + \int_0^x (t-x) \sin t \, dt$ for $x \in [0, \pi]$ if M and m are the maximum and minimum values of

$f(x)$ respectively in $[0, \pi]$ then $(M - m)$ equals

A) $\frac{1}{2}$

B) $\frac{\pi}{2}$

C) π

D) 3π

43. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is Inscribed in a rectangle R whose sides are parallel to the co-ordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R. The eccentricity of the ellipse E_2 is

A) $\frac{1}{\sqrt{2}}$

B) $\frac{\sqrt{3}}{2}$

C) $\frac{1}{2}$

D) $\frac{3}{4}$

44. If $1, \alpha_1, \alpha_2, \dots, \alpha_{2009}$ are $(2009)^{\text{th}}$ roots of unity, then the value of $2008 \sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ equals :

A) 2009

B) 2008

C) 0

D) -2009

45. There are 'n' urns $u_1, u_2, u_3 \dots u_n$. Each urn contains $(2n+1)$ balls. The i^{th} urn contains $2i$ number of white balls. Let $p(ui)$ which is probability of selecting i^{th} urn is proportional to $i^2 + 2$. If we randomly select one of the urns and draw one ball and probability of ball being white be $P(A)$; then $\lim_{n \rightarrow \infty} P(A)$ is equal to
- A) $\frac{3}{4}$ B) $\frac{1}{4}$ C) $\frac{2}{3}$ D) $\frac{3}{16}$
46. If $\sin(\sin x + \cos x) = \cos(\cos x - \sin x)$, then the value of $\sin x$ can be
- A) $\frac{\pi}{6}$ B) $\frac{\sqrt{16 - \pi^2}}{4}$ C) $\frac{\pi}{8}$ D) $\frac{-\pi}{6}$
47. If the distance from the point $P(1,1,1)$ to the line passing through the points $Q(0,6,8)$ and $R(-1,4,7)$ is expressed in the form $\sqrt{\frac{p}{q}}$ where p, q are co-prime, then the value of $\frac{(p+q)(p+q-1)}{2}$ equals
- A) 4950 B) 5050 C) 5150 D) 5250
48. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?
- A) f is not differentiable at $x = 0$
- B) f is monotonic
- C) f is even
- D) f has an Extremum
49. If a circle of radius 3 units is touching the lines $\sqrt{3}y^2 - 4xy + \sqrt{3}x^2 = 0$ in the first quadrant, then the length of the chord of contact to the circle is
- A) $\frac{\sqrt{3}+1}{2}$ B) $\frac{\sqrt{3}+1}{\sqrt{2}}$ C) $3\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)$ D) $3\left(\frac{\sqrt{3}+1}{2}\right)$

50. If $2a + 2b + 3c = \frac{1}{5}$ and $a, b, c > 0$ then maximum value of term of Independent of x in the Expansion

of $\left(abx^{\frac{1}{2}} + cx^{-\frac{1}{3}} \right)^{25}$ is

- A) $25_{C_{10}}$ B) $25_{C_{15}} (35)^{25}$ C) $25_{C_{15}} \left(\frac{1}{35} \right)^{35}$ D) $25_{C_{10}} \left(\frac{1}{25} \right)^{35}$

SECTION - II

(PARAGRAPH TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**. **Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

Paragraph for Question Nos. 51 to 52

Suppose $f(x)$ and $g(x)$ are two continuous functions defined for $0 \leq x \leq 2$. Given

$$f(x) = \int_0^1 e^{x+t} f(t) dt \text{ and } g(x) = \int_0^1 e^{x+t} g(t) dt + x$$

51. The value of $g(0) - f(0)$ equals.

- A) $\frac{2}{3-e^2}$ B) $\frac{3}{e^2-2}$ C) $\frac{1}{e^2-1}$ D) 0

52. The value of $\frac{g(0)}{g(2)}$ equals :

- A) 0 B) $\frac{1}{3}$ C) $\frac{1}{e^2}$ D) $\frac{2}{e^2}$

Paragraph for Question Nos. 53 to 54

A bag contains 4 white, 3 black and 2 red balls. Balls are drawn from the bag one by one without replacement.

53. The probability that 4th ball is red is equal to

- A) $\frac{5}{36}$ B) $\frac{1}{12}$ C) $\frac{2}{9}$ D) $\frac{3}{11}$

54. The probability that 5th ball is black given that 4th ball is red is equal to

- A) $\frac{1}{8}$ B) $\frac{3}{8}$ C) $\frac{2}{11}$ D) $\frac{3}{11}$

Paragraph for Question Nos. 55 to 56

Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

55. The ratio of the areas of the triangles PQS and PQR, is

- A) 1 : 3 B) 1 : 2 C) 2 : 3 D) 3 : 4

56. The equation of circle touching the parabola at upper end of its latusrectum and passing through its vertex, is

- A) $2x^2 + 2y^2 - x - 2y = 0$ B) $2x^2 + 2y^2 + 4x - \frac{9}{2}y = 0$
 C) $2x^2 + 2y^2 + x - 3y = 0$ D) $2x^2 + 2y^2 - 7x + y = 0$

SECTION - III**(Matching List Type)**

This section contains four questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

57.

	Column-I		Column-II
P)	Let $f(x) = \cos\left(\tan^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)$ Then $f\left(\frac{1}{4}\right)$ is equal to	1)	7
Q)	A bag contains 4 white, 3 black and 2 red balls. Balls are drawn from the bag one by one without replacements. If probability of 4 th ball is red equals $\frac{p}{q}$ where p and q are relatively prime, then $(q-p)$ equals	2)	4
R)	The area bounded by the curve $y = x + \sin x$ and its inverse function between the ordinates $x=0$ to $x=2\pi$ is 4s. Then	3)	2

	the value of s is		
S)	Let A_i where $i = 1, 2, 3 \dots 12$ are the vertices of a regular dodecagon and G is its center . Let 's' be the number of straight lines that can be formed with these 13 points, 't' be number of triangles that can be formed and 'd' be number of diagonals in a dodecagon ; then the value of $\frac{s+t+d}{100}$	4)	1

	P	Q	R	S		P	Q	R	S
A)	4	1	2	3	B)	4	1	3	2
C)	1	4	2	3	D)	1	4	3	2

58.

	Column-I		Column-II
P)	If the coefficient of x^2 and x^3 in the expansion of $(1+8x+bx^2)(1-3x)^9$ in the power of x are equal , then $\frac{7b}{27}$ is	1)	5
Q)	The number of integral values of α for which the equation $x^2 + 2(\alpha - 1)x + \alpha + 5 = 0$ has both roots positive and distinct is	2)	3
R)	Let $I = \int_0^1 (1-x^{50})^{99} x^{100} dx$ and	3)	2

	$J = \int_0^1 (1-x^{50})^{100} x^{100} dx.$ If $\frac{I}{J}$ is equal to $\frac{m}{n}$ where m and n are co-prime then the value of $\frac{m-n-1}{20}$		
S)	The slopes of sides BC, CA, AB of triangle ABC whose orthocenter is origin are $-1, -2, -3$ respectively. If locus of centroid of triangle ABC is $y = \left(\frac{m}{n}\right)x$. Where m,n are relatively prime then n-m is	4)	1

	P	Q	R	S
A)	3	2	1	4
B)	3	2	4	1
C)	3	2	1	1
D)	3	2	4	4

59.

	Column-I		Column-II
P)	A tangent to the hyperbola $\frac{x^2}{4} - y^2 = 1$ meets the ellipse $x^2 + 4y^2 = 4$ at two distinct points . The locus of mid-point of this chord is $(x^2 + 4y^2)^2 = k(x^2 - 4y^2)$, then 'k' is equal to	1)	2
Q)	Let the equation of the plane containing the line $x - y - z - 4 = 0 = x + y + 2z - 4$ and is parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$, then the value of $A + B + C$	2)	11
R)	Let ABC be a triangle whose centroid G, orthocenter is H and circumcentre is the origin O. If 'D' is any point in the plane of the triangle such that no three of O, A,B,C and D are collinear satisfying the relation $\vec{AD} + \vec{BD} + \vec{CH} + 3\vec{HG} = \lambda\vec{HD}$ then value of the scalar $\lambda + 4$ is	3)	4

S)	If $\alpha = \sin \left(\cot^{-1} \left(\tan \left(\cos^{-1} \left(\frac{2}{3} \right) \right) \right) \right)$ and $\beta = \sin \left(\cos e e^{-1} \left(\cot \left(\tan^{-1} \left(\frac{1}{3} \right) \right) \right) \right)$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ where a,b,c are integers and 'c' is prime then the value of $(a+b+c)$ equals	4)	6
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	P	Q	R	S
A)	2	1	4	3
B)	1	2	4	3
C)	3	2	4	1
D)	3	4	1	2

60. 6 married couples are present in a birth day party number of ways in which four persons are selected if :

	Column-I		Column-II
P)	They form no pair	P)	15
Q)	They form at least one pair	Q)	240
R)	They form at most one pair	R)	255
S)	They form exactly two pair	S)	480

	P	Q	R	S		P	Q	R	S
A)	2	4	3	1	B)	2	3	1	4
C)	3	2	4	1	D)	2	3	4	1

KEY SHEET

01	C	02	D	03	B	04	A	05	D	06	A
07	D	08	C	09	A	10	A	11	B	12	C
13	A	14	D	15	A	16	C	17	A	18	B
19	D	20	C								
21	A	22	A	23	C	24	A	25	D	26	D
27	A	28	C	29	C	30	A	31	B	32	A
33	B	34	B	35	D	36	A	37	D	38	C
39	C	40	B								
41	A	42	C	43	C	44	D	45	A	46	B
47	A	48	B	49	C	50	C	51	A	52	B
53	C	54	B	55	C	56	D	57	B	58	C
59	C	60	D								